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COA

CAPSTONE ASSIGNMENT

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BTech CSE FSD

Case Study 1 : Smart Traffic Control System

1) Role of basic functional Units

- Input Unit : Collects data from traffic sensors (vehicles, signals, cameras)
- memory Unit : Stores sensor data ~~into~~, control algorithms, and traffic rules.
- ALU (Arithmetic Logic Unit): Process data and makes decisions (eg, signal timing)
- Control Unit :- Coordinats all operations and manages real time responses
- Output Unit : sends commands to traffic lights and display systems.

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2) RISC Vs CISC Architecture

A RISC (Reduced Instruction Set computer) architecture is more suitable because:-

- faster execution due to simple instructions
- Better for real time processing
- Lower power consumption
- Easier pipelining compared to CISC

3) Instruction Throughput (IPC) and Performance

- Higher IPC (Instructions Per Cycle) improves system responsiveness
- Faster processing ensures quick reaction to traffic changes, reducing congestion
- Good performance is critical for real-time decision-making

4) Processor Type Selection
A pipelined processor is the best choice because:

- Executes multiple instructions simultaneously
- Improves efficiency and speed
- Suitable for continuous data processing in real-time systems.

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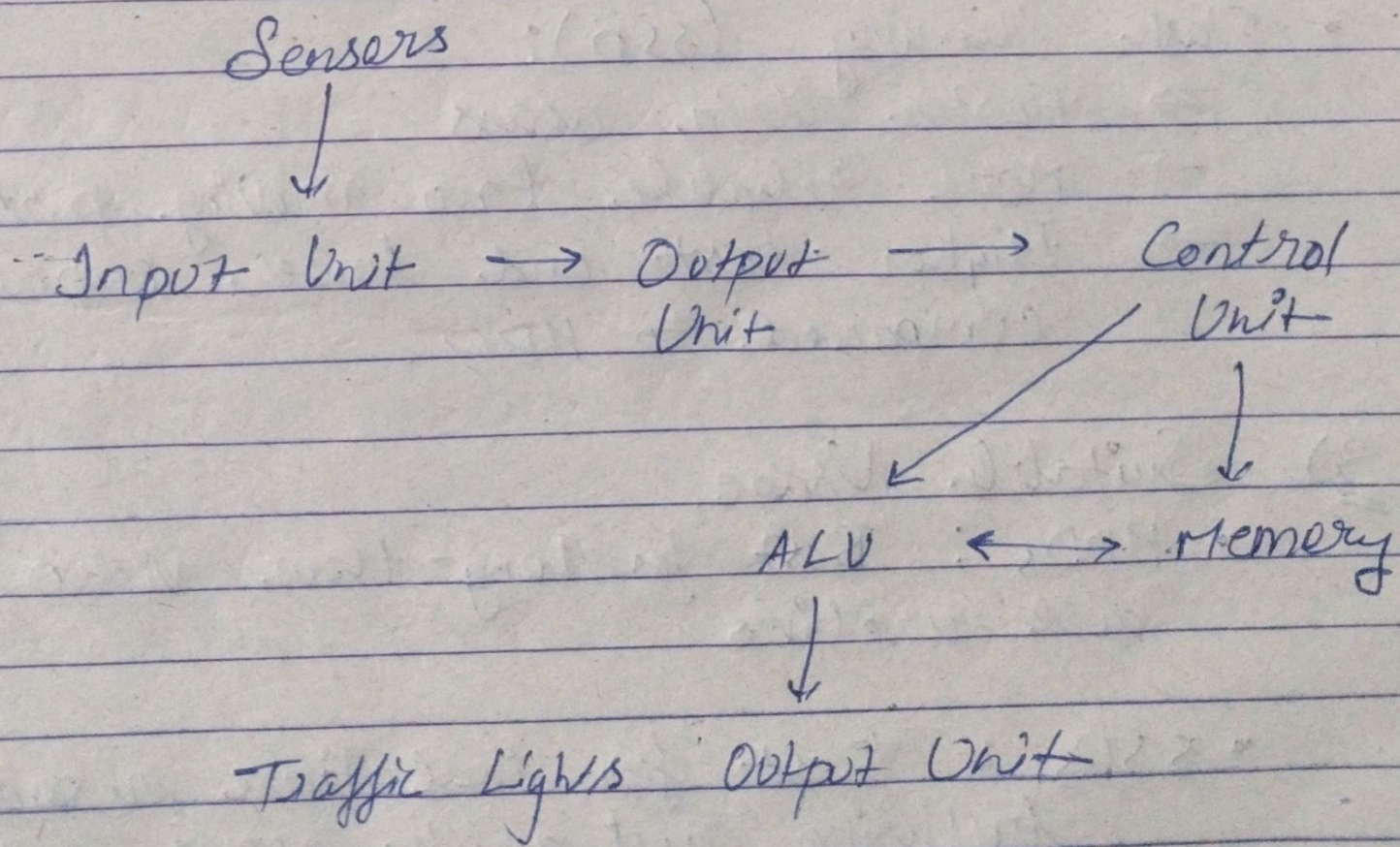
5) Pipeline Hazards and Solutions

- Data Hazards : Occur due to dependency between instructions.
  → Solution : forwarding, stalling

- Control Hazards : Due to branch Instructions
  → Solution : Branch prediction.

- Structural hazards : Resource conflicts
  → Solution : Better hardware design or resource duplication.

⇒ Conclusion :- A Risc - based pipelined processor with high IPC is ideal for a smart traffic control system due to its speed, efficiency, and ability to handle real-time operations reliably.



Flow Diagram

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Case 2 : Storage Planning for a Video Surveillance System

1) Role of Secondary Storage
Secondary storage is used to store large volumes of video data continuously. It provides high capacity, long term storage, and ensures recorded footage can be retrieved later for security analysis.

2) Magnetic Disks vs Flash Memory

- Magnetic Disks (HDD):
  → High storage capacity
  → Low cost per GB
  → Slower access speed.

- Flash Memory (SSD):
  → Faster data access
  → More reliable (no moving parts)
  → Higher cost and lower capacity compared to HDD.

3) Suitable Usage.
- HDDs : Best for long-term video storage and archives

- SSDs : Best for real-time recording, buffering and quick access data

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4) Memory Hierarchy and Retrieval Speed.
Memory hierarchy (cache - RAM - SSD → HDD) affects speed:

- Data stored in faster memory (cache/SSD), is accessed quickly
- Data in HDD takes more time.
- Frequently accessed video should be kep in faster storage to improve performance.

5) Optimized Architecture
Use a hybrid storage system:
- SSD layer : For live recording and recent footage
- HDD layer : for long-term storage.

optimally include cache memory for frequent accessed clips.

This balances cost (HDD), speed (SSD), and capacity (HDD + SSD combination)

⇒ Conclusion :-
A hybrid architecture using SSDs and HDDs is ideal, ensuring fast access, large storage capacity, and cost efficiency for a video survillance system

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Camera Input

↓

RAM / Buffer
(Temporary Storage)

↓

optional
↑
Cache Memory
Frequent
Access

SSD
(Fast Access layer
- Live Recording
→ Recent Footage

→

HDD
long-term Storage
→ Archives
→ old videos